

APEXPLANET TASK 2 — EDA REPORT

Exploratory Data Analysis & Business Intelligence

ApexPlanet 60-Day Data Analytics Internship
Cleaned Sales Dataset • 1,000 Records

1. Project Overview

This report documents the Exploratory Data Analysis and Business Intelligence work completed for ApexPlanet Task 2 using the cleaned sales dataset.

2. Objective

The objective is to uncover patterns, trends and relationships in the sales data and translate the analysis into practical business insights.

3. Dataset Overview

The cleaned dataset contains 1,000 records and 12 columns: Order_ID, Order_Date, Customer_ID, Customer_Name, Age, Gender, City, Product, Category, Quantity, Unit_Price and Total_Sales.

4. Data Quality Summary

- 1,000 records and 12 columns.
- No missing values were identified.
- No duplicate records were identified.
- Order_Date is datetime; numerical measures are numeric.
- 13 City records are labelled Unknown.

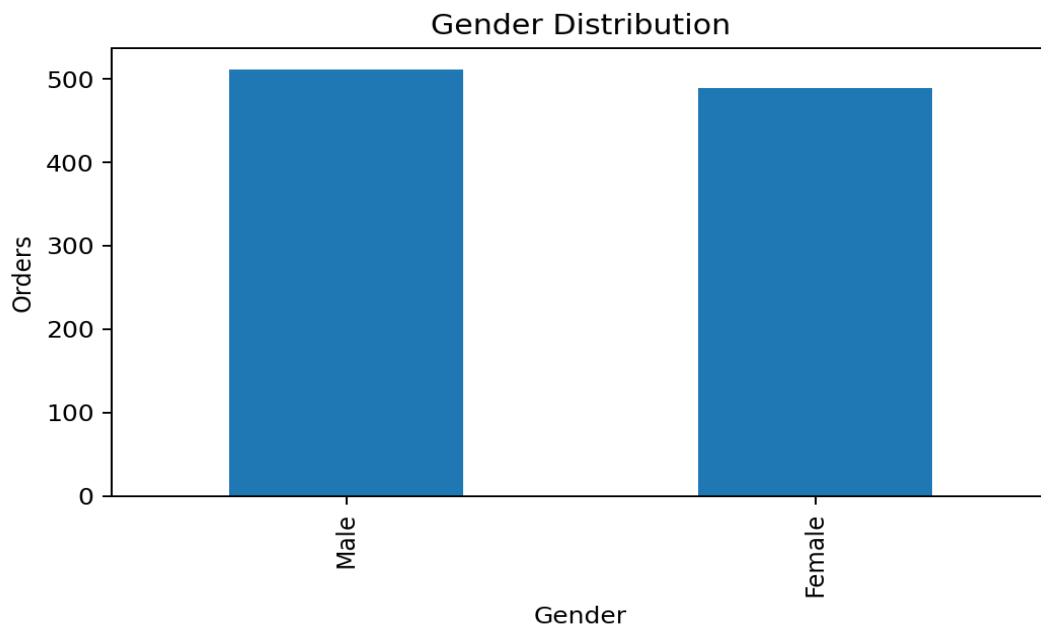
5. Descriptive Statistics

Variable	Mean	Median	Min	Max	Std. Dev.
Age	41.34	41.00	18.00	65.00	13.68
Quantity	5.43	5.00	1.00	10.00	2.84
Unit_Price	25,486.78	25,398.74	145.78	49,997.53	14,179.40
Total_Sales	139,399.44	108,594.02	437.34	493,677.50	114,100.05

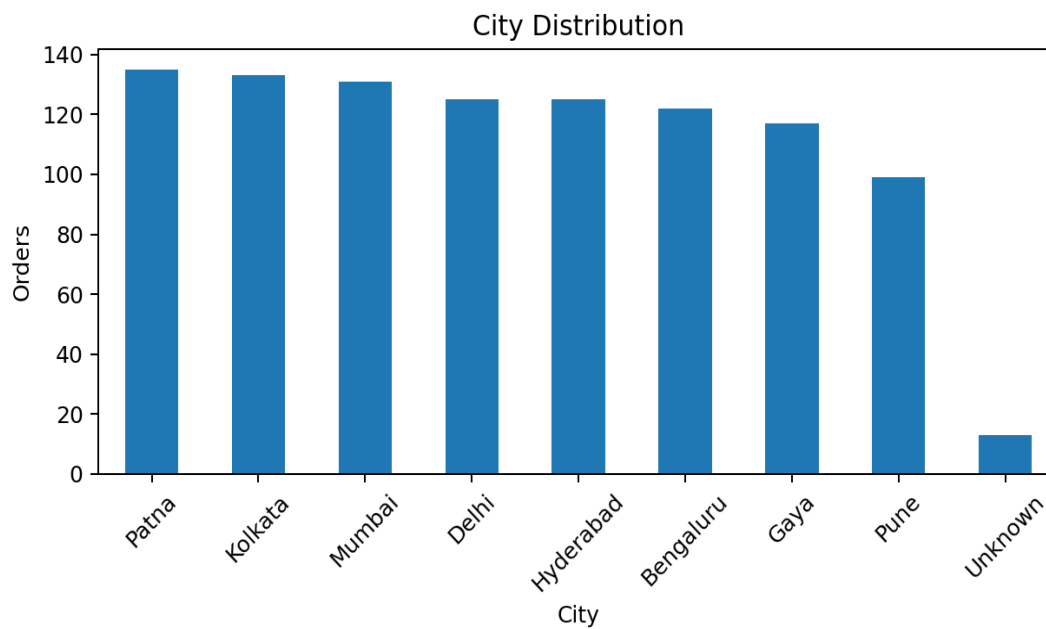
Average Total Sales per record is ■139,399.44 and average quantity is 5.44 units.

6. Univariate Analysis

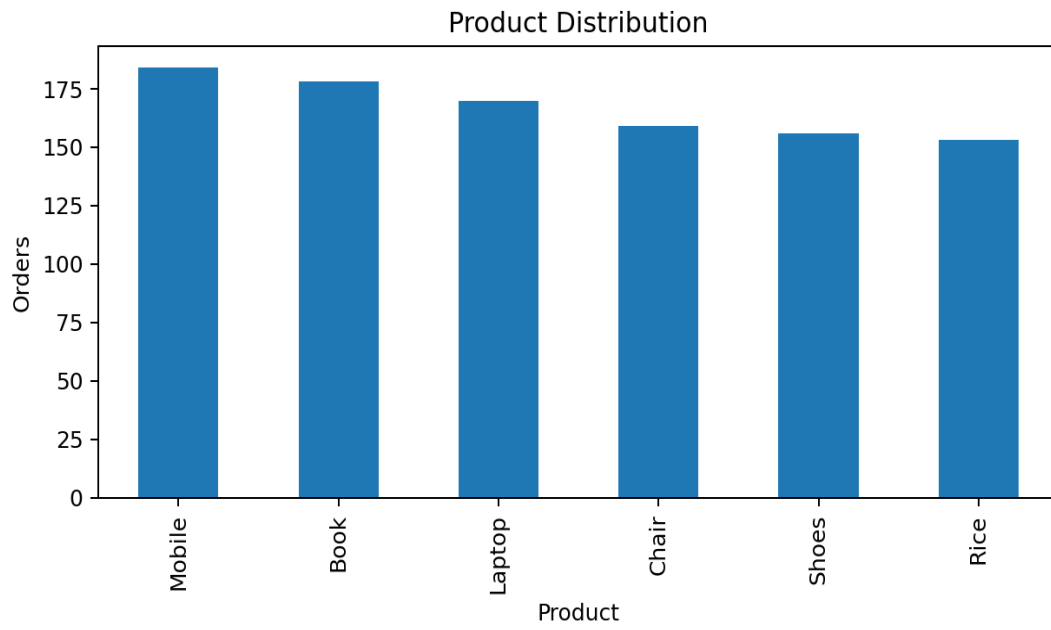
Eight visualizations examine gender, city, product, category, age, quantity, unit price and total sales.



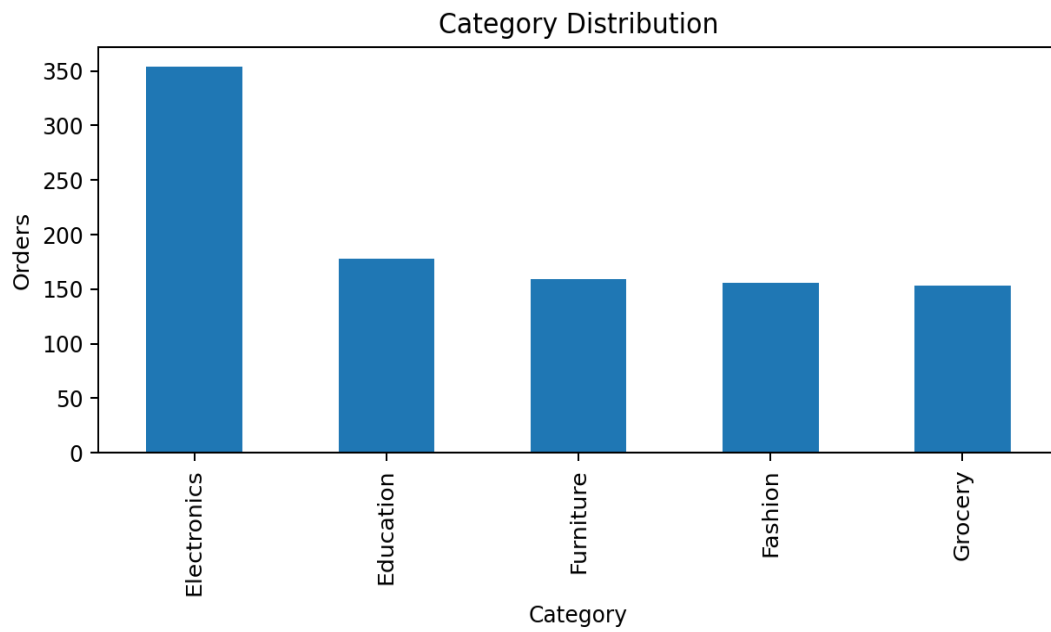
Gender Distribution



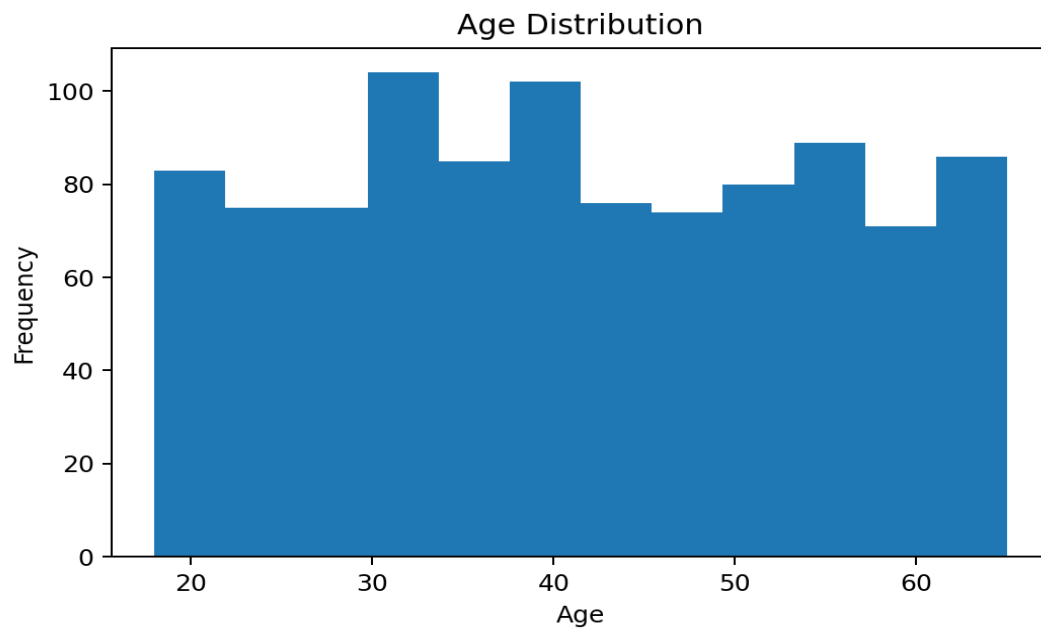
City Distribution



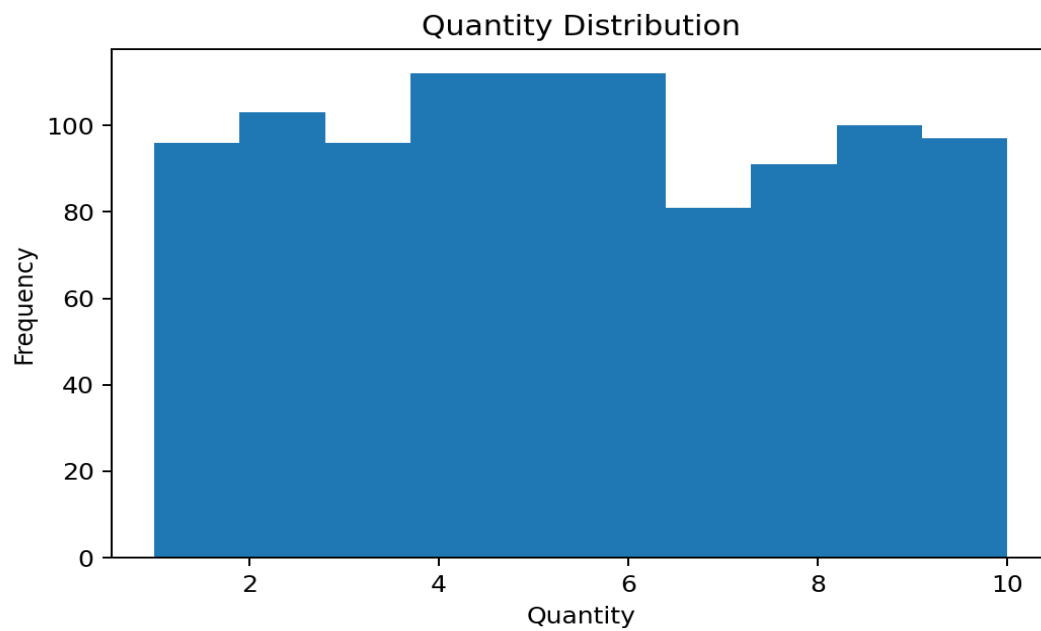
Product Distribution



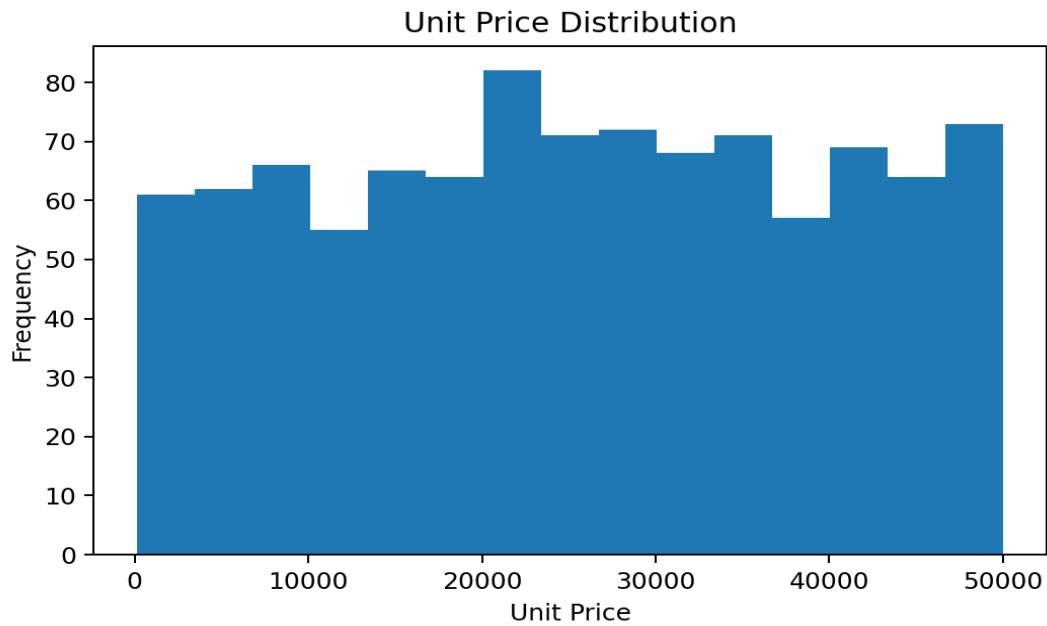
Category Distribution



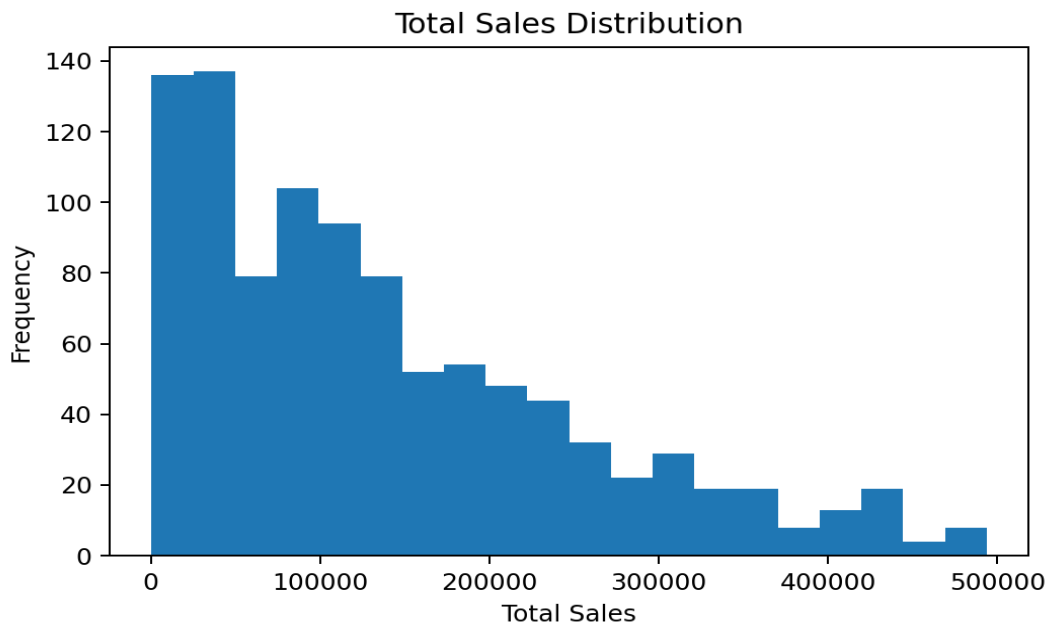
Age Distribution



Quantity Distribution



Unit Price Distribution



Total Sales Distribution

7. Categorical Analysis & Business Insights

- Mobile — 184 orders.
- Electronics — 354 orders.
- Patna — 135 orders.
- Male — 511 orders; Female — 489 orders.
- Average Total Sales — ■139,399.44.
- Average quantity — approximately 5.43 units.
- Highest single-order Total Sales — ■493,677.50.

8. SQL Business Analysis

Seven SQL business questions were prepared using filtering, aggregation and joins. Detailed SQL scripts and outputs are maintained in 03_SQL_Analysis.

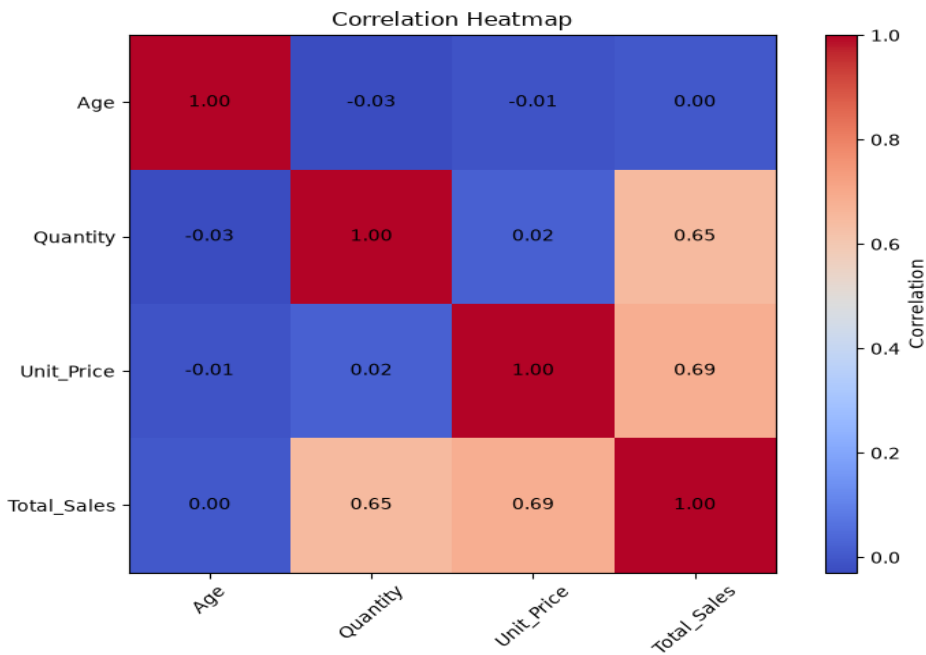
- Q1. City-wise order count

- Q2. Product-wise total quantity
- Q3. Category-wise total sales
- Q4. Average sales by city
- Q5. Category-wise quantity and sales
- Q6. Average sales by gender
- Q7. High-value orders above ₹100,000 by product/category

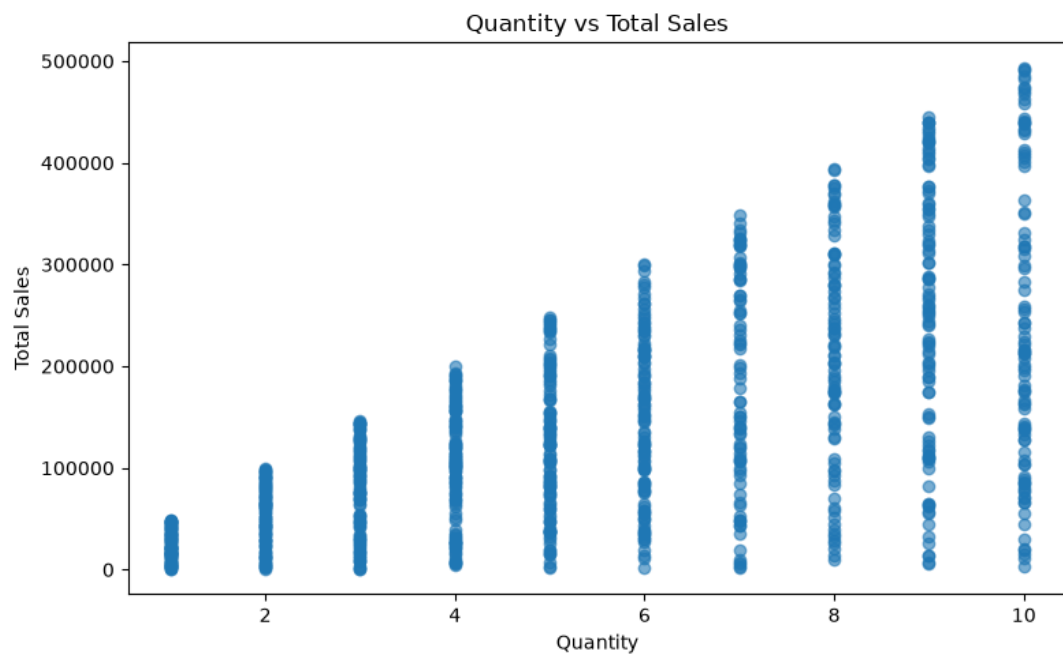
9. Multivariate Analysis & Correlation

Correlation describes association, not causation.

Variable Pair	Correlation	Interpretation
Age vs Quantity	-0.03	Almost no linear relationship
Age vs Unit_Price	-0.01	Almost no linear relationship
Age vs Total_Sales	0.00	Almost no linear relationship
Quantity vs Unit_Price	0.02	Almost no linear relationship
Quantity vs Total_Sales	0.65	Positive relationship
Unit_Price vs Total_Sales	0.69	Positive relationship



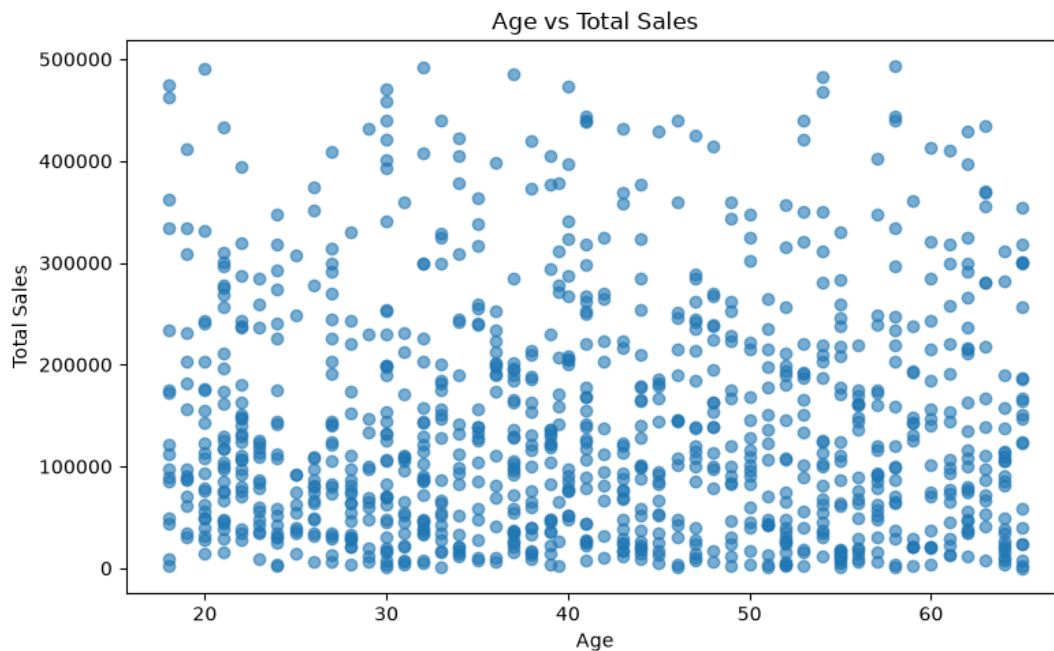
Correlation Heatmap



Quantity vs Total Sales



Unit Price vs Total Sales



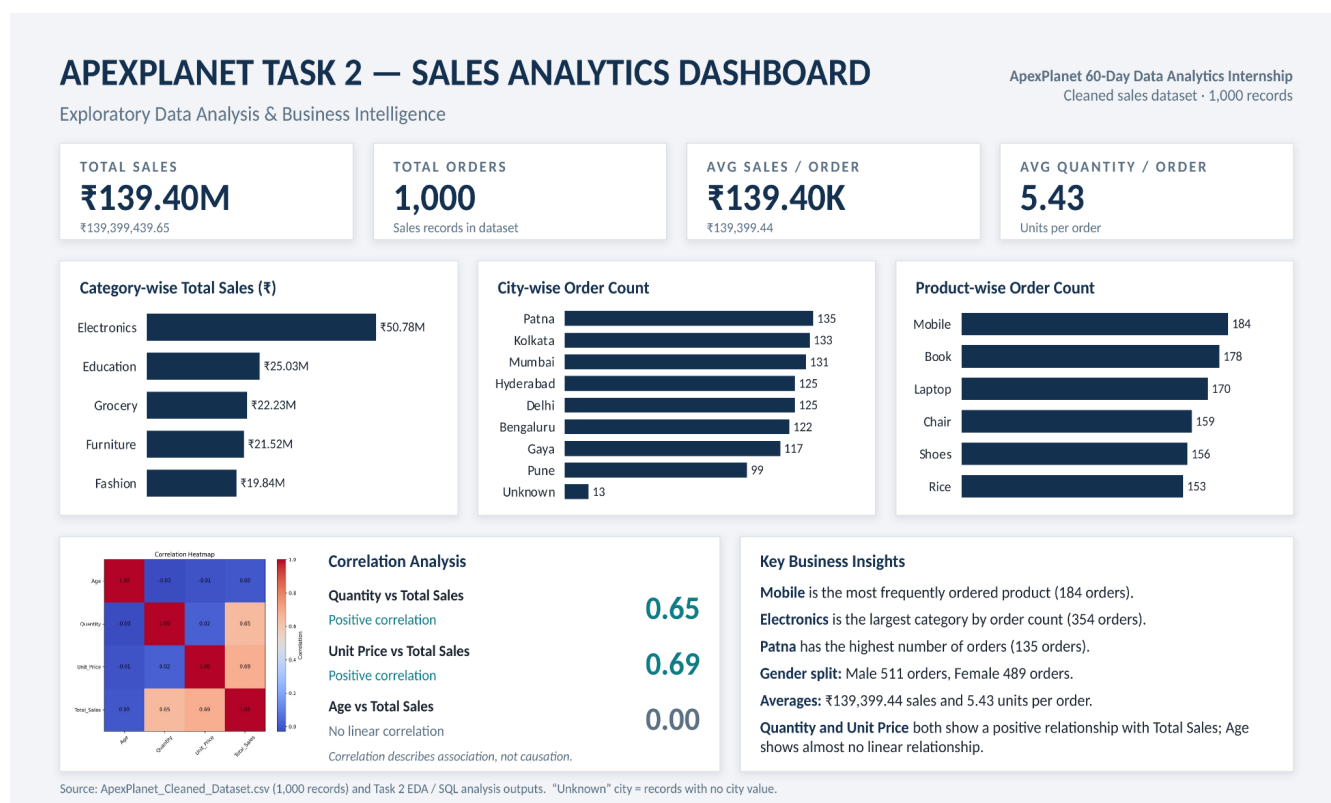
Age vs Total Sales

10. Key Findings

- Mobile has the highest product order count.
- Electronics has the highest category order count.
- Patna has the highest city order count.
- Quantity and Unit Price have positive relationships with Total Sales.
- Age has almost no linear relationship with Total Sales.

11. Static Dashboard Mock-up

A one-slide static Sales Analytics Dashboard was created as the visual summary for Task 2.



12. Conclusion

The Task 2 workflow covers data quality, descriptive statistics, univariate analysis, SQL business questions, multivariate analysis, correlation and dashboard communication.

13. Project Deliverables

- 01_Dataset
- 02_Python_EDA
- 03_SQL_Analysis
- 04_Visualizations
- 05_Dashboard_Mockup
- 06_EDA_Report